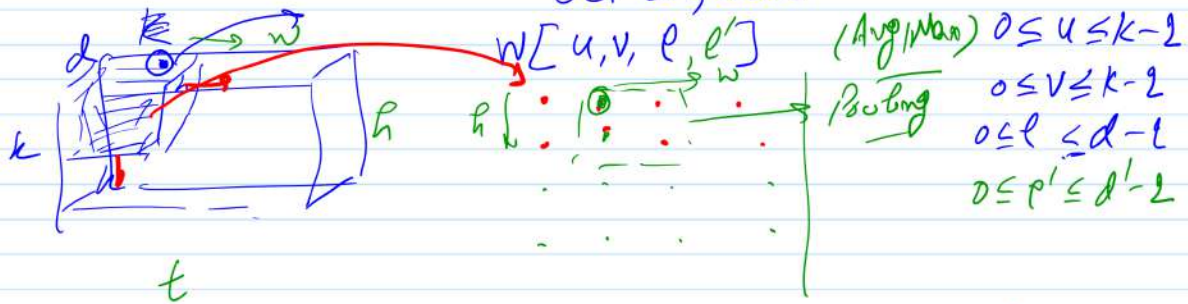


COL774 | COL7541

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$$0 \leq i \leq k-1$$

$$0 \leq j \leq d-1$$

$$z^{(t)}[i, d, p'] \xrightarrow{\text{kernel size}} = \sum_{u=0}^{k-1} \sum_{v=0}^{d-1} z^{(t)}[i+u, j+v, t] \cdot w[u, v, t, t'] + b[p']$$

$$z^{(t)}[i, d, p'] \rightarrow z^{(t+1)}[i, d, p']$$

$$\hat{z}^{(t+1)}[i, d, p'] = g(z^{(t+1)}[i, d, p'])$$

$$z^{(t+1)}[i, d, p'] = \frac{1}{b^2} \sum_{a,b=0}^{k-1} z^{(t+1)}[i+a, j+b, p']$$

pooling operation

$$\frac{\partial J}{\partial z^{(t+1)}[i, d, p']} = ? \sum_{i,j} \frac{\partial J}{\partial z^{(t)}[i, d, p']} \cdot \frac{\partial z^{(t)}[i, d, p']}{\partial w^{(t)}[u, v, t, t']}$$

$$\frac{\partial J}{\partial z^{(t)}[i, d, p']} = ? \sum_{u,v=0}^{k-1} \frac{\partial J}{\partial z^{(t)}[i-u, j-v, t']} \cdot \frac{\partial z^{(t)}[i-u, j-v, t']}{\partial w^{(t)}[u, v, t, t']}$$

H.W. $\frac{\partial \hat{z}^{(t+1)}[i, d, p']}{\partial z^{(t)}[i, d, p']}$

derivative of pooling operation

$$w[u, v, t, t']$$